25.4: The rise and fall of groups of organisms reflect differences in speciation and extinction rates

From its beginnings, life on Earth has been marked by the rise and fall of groups of organisms. Anaerobic prokaryotes originated, flourished, and then declined as the oxygen content of the atmosphere rose. Billions of years later, the first tetrapods emerged from the sea, giving rise to several major new groups of organisms. One of these, the amphibians, went on to dominate life on land for 100 million years, until other tetrapods (including dinosaurs and, later, mammals) replaced them as the dominant terrestrial vertebrates.

The rise and fall of these and other major groups of organisms have shaped the history of life. Narrowing our focus, we can also see that the rise or fall of any particular group is related to the speciation and extinction rates of its member species **(Figure 25.14)**. Just as a population increases in size when there are more births than deaths, the rise of a group of organisms occurs when more new species are produced than are lost to extinction. The reverse occurs when a group is in decline. In the **Scientific Skills Exercise**, you can interpret data from the fossil record about changes in a group of snail species in the early Paleogene period. Such changes in the fates of groups of organisms have been influenced by large-scale processes such as plate tectonics, mass extinctions, and adaptive radiations.

Figure 25.14

How speciation and extinction affect diversity.

Hypothetical tree diagram for the lineages of species A and B from over 4 million years ago to present.

The common ancestor of lineages A and B is at 4 million years ago. Lineage A splits around 3.2 million years ago; one branch splits around 2.2 million years ago, then each line ends within 500 million years; the other branch splits around 2.4 million years ago, then each line splits again around 1.2 to 1.3 million years ago. One branch splits and each part ends soon after, and the other branch splits and only one line ends. Lineage B splits around 3 million years ago. Each new branch splits between 2 and 2.4 million years ago. One branch goes to 2 lines that each have a branch; all 4 lines continue to present day. The other branch at 2 million years ago splits multiple times. Four out of 5 total lines continue to the present day.

The species diversity of an evolutionary lineage will increase when more new member species originate than are lost to extinction. In this hypothetical phylogenetic tree, each horizontal branch represents one species. By 2 million years ago both lineage A and lineage B have given rise to four species, and no species have

become extinct (denoted by a dagger symbol, †). By time 0, however, lineage A contains only one species, while lineage B contains eight species.

VISUAL SKILLS *Consider the period from* 2 million years *ago to time* 0*. For each lineage, determine how many speciation and extinction events occurred during that time.*

For suggested answer, see *Appendix A*.*

Scientific Skills Exercise

Estimating Quantitative Data from a Graph and Developing Hypotheses

AP Science Practices 2, 3, 5

Do Ecological Factors Affect Evolutionary Rates?Researchers studied the fossil record to investigate whether differing modes of larval dispersal might explain species longevity within one taxon of marine snails, the family Volutidae. Some of the snail species had nonplanktonic larvae: They developed directly into adults without a swimming stage. Other species had planktonic larvae: They had a swimming stage and could disperse very long distances. The adults of these planktonic species tended to have broad geographic distributions, whereas nonplanktonic species tended to be more isolated.

An assortment of shells.

How the Research Was Done The researchers studied the stratigraphic distribution of volutes in outcrops of sedimentary rocks located along North America's Gulf coast. These rocks, which formed from 66 to 37 million years ago, early in the Paleogene period, are an excellent source of well-preserved snail fossils. The researchers were able to classify each fossil species of volute snail as having planktonic or nonplanktonic larvae based on features of the earliest formed whorls of the snail's shell. Each bar in the graph shows how long one species of snail persisted in the fossil record.

Bar graph of snail species persistence in the fossil record with or without planktonic larva. Overall, species with planktonic larvae persisted much longer in the fossil record than species without.

Bar graph of how long snail species with or without planktonic larva persisted in the fossil record (horizontal axis is millions of years ago). Horizontal axis is millions of years ago (mya) ranging from 65 million years ago to 35 million years ago with increments ever 5 million years. From just over 65 to just over 55 million years

ago was the Paleocene; from just over 55 million years ago to 35 million years ago (and beyond) was the Eocene. Bars depicting how long snail species persisted in the fossil record are grouped by (1) species with planktonic larvae and (2) species without planktonic larva.

Planktonic Larvae	Appear	Extinct
Presence	(Millions of years ago, values approximate)	(Millions of years ago, values approximate)
Present	63	57
Present	63	57
Present	63	57
Present	59	58
Present	58	56
Present	58	51
Present	51	50
Present	51	41
Present	47	45
Present	47	43
Present	47	43
Present	47	40
Present	43	38
Absent	64	63
Absent	64	63
Absent	64	59
Absent	59	89
Absent	58	55

Absent	58	55
Absent	56	55
Absent	47	46
Absent	47	46
Absent	47	13
Absent	46	43
Absent	46	43
Absent	45	43
Absent	45	43
Absent	45	43
Absent	45	43
Absent	43	42
Absent	43	42
Absent	43	42
Absent	43	42
Absent	43	42
Absent	43	42
Absent	43	42
Absent	41	40
Absent	41	40
Absent	41	40
Absent	41	40
Absent	41	39

Data from T. A. Hansen, Larval dispersal and species longevity in Lower Tertiary gastropods, *Science* 199:885–887 (1978). Reprinted with permission from AAAS.

INTERPRET THE DATA

- You can estimate quantitative data (fairly precisely) from a graph. The first step is to obtain a conversion factor by measuring along an axis that has a scale. In this case, 25 million years (my; from 60 to 35 million years ago [mya] on the x-axis) is represented by a distance of 7.0 cm. This yields a conversion factor (a ratio) of 25my/7.0 cm=3.6my/cm. To estimate the time period represented by a horizontal bar on this graph, measure the length of that bar in centimeters and multiply that measurement by the conversion factor, 3.6my/cm. For example, a bar that measures 1.1 cm on the graph represents a persistence time of 1.1 cm×3.6my/cm=4million years.
- Calculate the mean (average) persistence times for species with planktonic larvae and species with nonplanktonic larvae.
- Count the number of new species that form in each group beginning at 60 mya (the first three species in each group were present around 64 mya, the first time period sampled, so we don't know when those species first appear in the fossil record).
- Propose a hypothesis to explain the differences in longevity of snail species with planktonic and nonplanktonic larvae.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Plate Tectonics

If photographs of Earth were taken from space every 10,000 years and spliced together to make a movie, it would show something many of us find hard to imagine: The seemingly "rock solid" continents we live on move over time. Over the past billion years, there have been three occasions (1 billion, 600 million, and 250 million years ago) when most of the landmasses of Earth came together to form a supercontinent, then later broke apart. Each time, this breakup yielded a different configuration of continents. Based on the directions in which the continents are moving today, some geologists have estimated that a new supercontinent will form roughly 250 million years from now.

According to the theory of **plate tectonics**, the continents are part of great plates of Earth's crust that essentially float on the hot, underlying portion of the mantle **(Figure 25.15)**. Movements in the mantle cause the plates to move over time in a process called *continental drift*. Geologists can measure the rate at which the plates are moving now, usually only a few centimeters per year. They can also infer the past locations of the continents using the magnetic signal recorded in rocks at the time of their formation. This

method works because as a continent shifts its position over time, the direction of magnetic north recorded in its newly formed rocks also changes.

Figure 25.15

Cutaway view of Earth.

Layers of the earth from the outside in: thin crust, followed by thick mantle, and then an outer core about two thirds as thick, followed by thinner inner core.

The thickness of the crust is exaggerated here.

Earth's major tectonic plates are shown in **Figure 25.16**. Many important geologic processes, including the formation of mountains and islands, occur at plate boundaries. In some cases, two plates are moving away from each other, as are the North American and Eurasian plates, which are currently drifting apart at a rate of about 2 cm per year. In other cases, two plates are sliding past each other, forming regions where earthquakes are common. California's infamous San Andreas Fault is part of a border where two plates slide past each other. In still other cases, two plates collide, producing violent upheavals and forming new mountains along the plate boundaries. One spectacular example of this occurred 45 million years ago, when the Indian plate crashed into the Eurasian plate, starting the formation of the Himalayan mountains.

Figure 25.16

Earth's major tectonic plates.

Map of the world's continents with direction of tectonic plate movement. 15 plates are labeled, with regions of violent tectonic activity highlighted at points where some plates intersect.

The blue arrows indicate direction of movement. The reddish orange dots represent zones of violent tectonic activity.

HHMI BioInteractive Animation

Plate Tectonics

image

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Consequences of Continental Drift

Plate movements rearrange geography slowly, but their cumulative effects are dramatic. In addition to reshaping the physical features of our planet, continental drift also has a major impact on life on Earth.

One reason for this is that continental drift alters the habitats in which organisms live. Consider the changes shown in **Figure 25.17**. About 250 million years ago, plate movements brought previously separated landmasses together into a supercontinent named **Pangaea**. Ocean basins became deeper, which drained shallow coastal seas. At that time, as now, most marine species inhabited shallow waters, and the formation of Pangaea destroyed much of that habitat. Pangaea's interior was cold and dry, probably an even more severe environment than that of central Asia today. Overall, the formation of Pangaea greatly altered the physical environment and climate, which drove some species to extinction and provided new opportunities for groups of organisms that survived the crisis.

Figure 25.17

The history of continental drift during the Phanerozoic eon.

Positions of continents shift from Pangaea in the Paleozoic (one large continent) to Laurasia and Gondwana in the Mesozoic until they break up into present formation.

At the end of the Paleozoic (251 million years ago), all of the Earth's landmasses were joined in the supercontinent Pangaea which spanned the equator. By the mid-Mesozoic (135 million years ago), Pangaea split into northern (Laurasia) and southern (Gondwana) landmasses. By the end of the Mesozoic (65.5 million years ago), Laurasia and Gondwana separated into the present-day continents. North America, South America, Eurasia, Africa, Antarctica, India and Madagascar are all regions labeled on the map. Earth's youngest major mountain range, the Himalayas, began to form when India collided with Eurasia about 45 million years ago (in the Cenozoic). The continents continue to drift today.

VISUAL SKILLS *Is the Australian plate's current direction of movement (see*Figure 25.16*)* similar to the direction it traveled over the past*66 million years*?*

*For suggested answer, see*Appendix A*.*

Organisms are also affected by the climate change that results when a continent shifts its location. The southern tip of Labrador, Canada, for example, once was located in the tropics but has moved 40° to the north over the last 200 million years. When faced with the changes in climate that such shifts in position entail, organisms adapt, move to a new location, or become extinct (this last outcome occurred for many organisms stranded on Antarctica, which separated from Australia 40 million years ago).

Continental drift also promotes allopatric speciation on a grand scale. When supercontinents break apart, regions that once were connected become isolated. As the continents drifted apart over the last 200 million years, each became a separate evolutionary arena, with lineages of plants and animals that diverged from those on other continents. For example, marsupial mammals fill ecological roles in Australia analogous to those filled by eutherians (placental mammals) on other continents (see Figure 22.18). Fossil evidence shows that marsupial mammals originated in what is now Asia and reached Australia via South America and Antarctica while the continents were still joined. The subsequent breakup of the southern continents set Australia "afloat," like a giant raft of marsupials. In Australia, marsupials diversified, and the few eutherians (placental mammals) that lived there became extinct; on other continents, most marsupials became extinct and the eutherians diversified.

Finally, continental drift can help explain puzzling geographic distributions of extinct organisms, such as fossils of the same species of Permian freshwater reptiles found in both Brazil and the West African nation of Ghana. These two parts of the world, now separated by 3,000 km of ocean, were joined together when these reptiles were living.

HHMI BioInteractive Video: Animated Life

Pangea, Wegener, and Continental Drift

image

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Mass Extinctions

The fossil record shows that the overwhelming majority of species that ever lived are now extinct. A species may become extinct for many reasons. Its habitat may have been destroyed, or its environment may have changed in a manner unfavorable to the species. For example, if ocean temperatures fall by even a few degrees, species that are otherwise well adapted may perish. Even if physical factors in the environment remain stable, biological factors may change—the origin of one species can spell doom for another.

Although extinction occurs regularly, at certain times disruptive changes to the global environment have caused the rate of extinction to increase dramatically. The result is a **mass extinction**, in which large numbers of species become extinct worldwide.

The “Big Five” Mass Extinction Events

Five mass extinctions are documented in the fossil record over the past 500 million years**(**Figure 25.18**)**. These events are particularly well documented for the decimation of hard-bodied animals that lived in shallow seas, the organisms for which the fossil record is most complete. In each mass extinction, 50% or more of marine species became extinct.

Figure 25.18

Mass extinction and the diversity of life.

A graph shows that an increase in the number of marine animal families over time is kept in check by 5 peaks (mass extinction events), represented by a second line.

The five generally recognized mass extinction events, indicated by red arrows, represent peaks in the extinction rate of marine animal families (red line and left vertical axis). These mass extinctions interrupted the overall increase, over time, in the number of extant families of marine animals (blue line and right vertical axis).

Left vertical axis is total extinction rate (families per million years) ranging from 0 to 25 with increments every 5 million years. Right vertical axis is number of extant families ranging from 0 to 1,100 with increments every 100. Horizontal axis is time (millions of years ago) ranging from 541 to present, divided into the Paleozoic, Mesozoic and Cenozoic eras and periods represented with single letters (See data table). Extinction rates peak 444, 365, 252, 200 and 66 million years ago, marking the 5 mass extinction events. Corresponding total extinction rates (families per million years) for each mass extinction event are: 20; 10; 27.5; about 11; and about 16. General trends: Number of families increases to just under 500 about 444 million years ago, then drops following the first mass extinction event at that time to a little more than 350, then increases to 500 before dropping during the next mass extinction event (365 million years ago) to under 400. Number of families increases again to fluctuate around 450, until it drops to about 200 during the mass extinction event 252 million years ago, which represents the largest peak extinction rate on the graph at 27.5 families per million years (and therefore the largest total drop in number of families). Number of families then increases to about 300, dipping slightly during the mass extinction even 200 million years ago, before climbing rapidly to over 600 by 66 million years ago. This point marks the fifth mass extinction event, when number of families dips to just over 500, before climbing again into the present (just under 800 families).

INTERPRET THE DATA *As mentioned in the text,*96%*of marine animal species became extinct in the Permian mass extinction (252 million yearsago). Explain why the blue curve shows only a*50%*drop at that time.*

Two mass extinctions—the Permian and the Cretaceous—have received the most attention. The Permian mass extinction, which defines the boundary between the Paleozoic and Mesozoic eras (252 million years ago), claimed about 96% of marine animal species and drastically altered life in the ocean. Terrestrial life was also affected. For example, 8 out of 27 known orders of insects were wiped out. This mass extinction occurred in less than 500,000 years, possibly in just a few thousand years—an instant in the context of geologic time.

The Permian mass extinction occurred during the most extreme episode of volcanism in the past 500 million years. Geologic data indicate that 1.6 million km2 (roughly half the size of Western Europe) in Siberia was covered with lava hundreds of meters thick. The eruptions are thought to have produced enough carbon dioxide to warm the global climate by an estimated 6°C, harming many temperature-sensitive species. The rise in atmospheric CO_2 levels would also have led to ocean acidification, thereby reducing the availability of calcium carbonate, which is required by reef-building corals and many shell-building species (see Figure 3.12). The eruptions would also have added nutrients such as phosphorus to marine ecosystems, stimulating the growth of microorganisms. Upon their deaths, these microorganisms would have provided food for bacterial decomposers. Bacteria use oxygen as they decompose the bodies of dead organisms, thus causing oxygen concentrations to drop. This would have harmed oxygen-breathers and promoted the growth of anaerobic bacteria that emit a poisonous metabolic by-product, hydrogen sulfide (H_2S) gas. Overall, the volcanic eruptions may have triggered a series of catastrophic events that together resulted in the Permian mass extinction.

The Cretaceous mass extinction occurred 66 million years ago. This event extinguished more than half of all marine species and eliminated many families of terrestrial plants and animals, including all dinosaurs (except birds, which are members of the same group; see Figure 34.25). One clue to a possible cause of the Cretaceous mass extinction is a thin layer of clay enriched in iridium that dates to the time of the mass extinction. Iridium is an element that is very rare on Earth but common in many of the meteorites and other extraterrestrial objects that occasionally fall to Earth. As a result, researchers proposed that this clay is fallout from a huge cloud of debris that billowed into the atmosphere when an asteroid collided with Earth. This cloud would have blocked sunlight and caused a sudden drop in global temperatures lasting for several months to years.

Is there evidence of such an asteroid? Research has focused on the Chicxulub crater, a 66-million-year-old scar beneath sediments off the coast of Mexico **(Figure 25.19)**. The crater is the right size to have been caused by an object with a diameter of 10 km. When the impact occurred, many species may have been particularly vulnerable to extinction because recent large-scale volcanic eruptions had already caused their populations to decline. Moreover, a 2018 study estimated that wildfires resulting from the impact contributed to a sharp rise in atmospheric CO_2 levels, causing a period of global warming that lasted

100,000 years—yet another stress for populations already in decline. Critical evaluation of these and other hypotheses for mass extinctions continues.

Figure 25.19

A trauma for Cretaceous life.

The Chicxulub crater is located on the Yucatan Peninsula of North America. An artist's rendering of the asteroid shows a massive impact.

Beneath the Caribbean Sea, the 66-million-year-old Chicxulub crater measures 180 km across. The horseshoe shape of the crater and the pattern of debris in sedimentary rocks indicate that an asteroid struck at a low angle. This drawing represents the impact and its immediate effect: a cloud of hot vapor and debris that could have killed many of the plants and animals in North America within hours and ignited massive wildfires.

HHMI BioInteractive Video

The Day the Mesozoic Died

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Is a Sixth Mass Extinction Under Way?

As you will read further in Concept 56.1, human actions, such as habitat destruction, are modifying the global environment to such an extent that many species are threatened with extinction. More than 1,000 species have become extinct in the last 400 years. Scientists estimate that this rate is 100 to 1,000 times the typical background rate seen in the fossil record. Is a sixth mass extinction now in progress?

This question is difficult to answer, in part because it is hard to document the total number of extinctions occurring today. Tropical rain forests, for example, harbor many undiscovered species. As a result, destroying tropical forest may drive species to extinction before we even learn of them.

Such uncertainties make it hard to assess the full extent of the current extinction crisis. Even so, it is clear that losses to date have not reached those of the "big five" mass extinctions, in which large percentages of Earth's species became extinct. This does not in any way discount the seriousness of today's situation.

Monitoring programs show that many species are declining at an alarming rate due to habitat loss, introduced species, overharvesting, and other factors. Recent studies on a variety of organisms, including lizards, pine trees, and polar bears, suggest that climate change may hasten some of these declines. The fossil record also highlights the potential importance of climate change: Over the last 500 million years, extinction rates have tended to increase when global temperatures were high **(Figure 25.20)**. Overall, the evidence suggests that unless dramatic actions are taken, a sixth, human-caused mass extinction is likely to occur within the next few centuries.

Figure 25.20

Fossil extinctions and temperature.

A graph plots relative extinction rate of marine animal genera versus relative temperature from negative 3 to 4.

A scatterplot generally rises from left to right with mass extinctions occurring toward the higher end where temperatures are warmer. One mass extinction occurred at approximately (negative 1.8, 2.1), which was during a period of cooler temperatures.

Extinction rates increased when global temperatures were high. Temperatures were estimated using ratios of oxygen isotopes and converted to an index in which 0 is the overall average temperature.

Consequences of Mass Extinctions

Mass extinctions have significant and long-term effects. By eliminating large numbers of species, a mass extinction can reduce a thriving and complex ecological community to a pale shadow of its former self. And once an evolutionary lineage disappears, it cannot reappear. The course of evolution is changed forever. Consider what would have happened if the early primates living 66 million years ago had died out in the Cretaceous mass extinction. Humans would not exist, and life on Earth would differ greatly from what it is today.

The fossil record shows that it typically takes 5–10 million years for the diversity of life to recover to previous levels after a mass extinction. In some cases, it has taken much longer than that: It took about 100 million years for the number of marine families to recover after the Permian mass extinction (see Figure 25.18). These data have sobering implications. If current trends continue and a sixth mass extinction occurs, it will take millions of years for life on Earth to recover.

Mass extinctions can also alter ecological communities by changing the types of organisms residing there. For example, after the Permian and Cretaceous mass extinctions, the percentage of marine organisms that were predators increased substantially **(Figure 25.21)**. A rise in the number of predators can increase both

the risks faced by prey and the competition among predators for food. In addition, mass extinctions can curtail lineages with novel and advantageous features. For example, in the late Triassic period, a group of gastropods (snails and their relatives) arose that could drill through the shells of bivalves (such as clams) and feed on the animals inside. Although shell drilling provided access to a new and abundant source of food, this newly formed group was wiped out during the mass extinction at the end of the Triassic (about 200 million years ago). Another 120 million years passed before another group of gastropods (the oyster drills) exhibited the ability to drill through shells. As their predecessors might have done if they had not originated at an unfortunate time, oyster drills have since diversified into many new species. Finally, by eliminating so many species, mass extinctions can pave the way for adaptive radiations, in which new groups of organisms proliferate.

Figure 25.21

Mass extinctions and ecology.

A graph plots Predator genera in percentage of marine genera versus time in millions of years ago.

The Permian and Cretaceous mass extinctions (indicated by red arrows) altered the ecology of the oceans by increasing the percentage of marine genera that were predators.

A jagged graph rises from (541, 2) to (480, 22), then rises and falls between 9% and 20% until 252 million years ago. During this time, the percentage of marine genera that were predators remained close to 15% for 200 million years. The graph spikes to (252, 28), then rises and falls in the low twenties until 66 million years ago. After the Permian mass extinction, the percentage of predators rose sharply then stabilized near 22% for 150 million years - until it was disrupted again by the Cretaceous mass extinction. The graph rises to approximately 36% for the present day.

Adaptive Radiations

The fossil record shows that the diversity of life has increased over the past 250 million years (see blue line in Figure 25.18). This increase has been fueled by **adaptive radiations**, periods of evolutionary change in which groups of organisms form many new species whose adaptations allow them to fill different ecological roles, or niches, in their communities. Large-scale adaptive radiations occurred after each of the big five mass extinctions, when survivors became adapted to the many vacant ecological niches. Adaptive radiations have also occurred in groups of organisms that possessed major evolutionary innovations, such as seeds or armored body coverings, or that colonized regions in which they faced little competition from other species.

Worldwide Adaptive Radiations

Fossil evidence indicates that mammals underwent a dramatic adaptive radiation after the extinction of terrestrial dinosaurs 66 million years ago **(Figure 25.22)**. Although mammals originated about 180 million years ago, the mammal fossils older than 66 million years are mostly small and show less morphological diversity than found today. Many species appear to have been nocturnal based on their large eye sockets, similar to those in living nocturnal mammals. A few early mammals were intermediate in size, such as *Repenomamus giganticus*, a 1-m-long predator that lived 130 million years ago—but none approached the size of many dinosaurs. Early mammals may have been restricted in size and diversity because they were eaten or outcompeted by the larger and more diverse dinosaurs. With the disappearance of the dinosaurs (except for birds), mammals expanded greatly in both diversity and size, filling the ecological roles once occupied by terrestrial dinosaurs.

Figure 25.22

Adaptive radiation of mammals.

An evolutionary tree.

The ancestral cynodont leads to the ancestral mammal at 170 million years ago. One branch goes to modern day monotremes (5 species). Another branch goes to 155 million years ago and splits. One part goes to marsupials (366 species), and the other part goes to eutherians (placental mammals; 6,028 species).

The history of life has also been greatly altered by radiations in which groups of organisms increased in diversity as they came to play entirely new ecological roles in their communities. As we'll explore in later chapters, examples include the rise of photosynthetic prokaryotes, the evolution of large predators in the Cambrian explosion, and the radiations following the colonization of land by plants, insects, and tetrapods. Each of these last three radiations was associated with major evolutionary innovations that facilitated life on land. The radiation of plants, for example, was associated with key adaptations, such as stems that support plants against gravity and a waxy coat that protects leaves from water loss. Finally, organisms that arise in an adaptive radiation can serve as a new source of food for still other organisms. In fact, the diversification of plants stimulated a series of adaptive radiations in insects that ate or pollinated plants, one reason that insects are the most diverse group of animals on Earth today.

Regional Adaptive Radiations

Striking adaptive radiations have also occurred over more limited geographic areas. Such radiations can be initiated when a few organisms make their way to a new, often distant location in which they face relatively little competition from other organisms. The Hawaiian archipelago is one of the world's great showcases of

this type of adaptive radiation **(Figure 25.23)**. Located about 3,500 km from the nearest continent, the volcanic islands are progressively older as one follows the chain toward the northwest; the youngest island, Hawaii, is less than a million years old and still has active volcanoes. Each island was born “naked” and was gradually populated by stray organisms that rode the ocean currents and winds either from far-distant land areas or from older islands of the archipelago itself. The physical diversity of each island, including immense variation in soil conditions, elevation, and rainfall, provides many opportunities for evolutionary divergence by natural selection. Multiple invasions followed by speciation events have ignited an explosion of adaptive radiation in Hawaii. As a result, thousands of species that inhabit the islands are found nowhere else on Earth. Among plants, for example, about 1,100 species are unique to the Hawaiian Islands. Unfortunately, many of these species are now facing an elevated risk of extinction due to human actions such as habitat destruction and the introduction of non-native plant species.

Figure 25.23

Adaptive radiation on the Hawaiian Islands.

Adaptive radiation on the Hawaiian Islands.

Five plants evolved from a close North American relative, the tarweed Carlquistia muirii. Dubautia waialealae evolved 5.1 million years ago on Kauai. Dubautia laxa evolved 3.7 million years ago on Oahu. Dubautia scabra evolved 0.4 to 1.3 million years ago on Molokai, Lanau, Maui, and the Big Island. Argyroxiphium sandwicense evolved 0.4 to 1.3 million years ago on Maui and the Big Island. Dubautia linearis evolved 0.4 million years ago on the Big Island.

Molecular analysis indicates that these remarkably varied Hawaiian plants, known collectively as the “silversword alliance,” are all descended from an ancestral tarweed that arrived on the islands from North America about 5 million years ago. Silverswords have since spread into different habitats and formed new species with strikingly different adaptations.

Animation

Overview of Macroevolution

image

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